Studio Resources

This guide has been adapted and simplified for CDE2501.

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Site Visit Guide

This guide has been adapted and simplified for CDE2501.

Purpose of this Guide

- A general guide for you to refer to when you visit your district.
- Includes some pointers that may help you make the most of your site visits by learning how to 'observe effectively' *(Musante and DeWalt, 2010).*
- This checklist, however, is not exhaustive.
- Disclaimer: adhering (strictly) to this checklist may risk you losing other observations; remember to keep an open mind when you do your fieldwork!

Introduction to Site visits and Observational study

A useful research method to:

- Identifying and describing phenomenon, and
- exploring and explaining how the phenomenon work (Vogt et al, 2012)

This guide will detail Cohen et al's (2018) checklist for researchers conducting site visits (modified for CDE2501). The following pages will bring you through each step with some prompts and guiding questions.

1. Locating a field of study
2. Deciding on sampling
3. Finding informants
4. Collecting data
5. Analysing data

Before Site Visit

1. **Research the district/site** – e.g. history, demographics, current issues through desktop research (refer to Research Methods Guide for more details).

2. From your desktop research, **identify potential themes and areas of interest** to look out for during the site visit – e.g. transportation, housing, green spaces, population ageing.

3. Before going down to site**, plan ahead your data collection method and design ahead your questions**.
   - Have a brief idea of what data you want to collect when you are at site. For example, if you decide to do an interview, identify who you want to interview and why (e.g. demographic profiles of interviewees), what kind of questions you want to ask
   - Tips: you can consult your SLs beforehand), and how many people you want to survey or interview (refer to Research Methods Guide for more details).

4. Choose equipment to bring along for site visit:
   - Notebook/pen or tablet/phone for notetaking/sketching
   - Phone: audio recording, taking pictures and videos, maps etc..
   - Umbrella for wet weather, hat and water bottle for hot weather etc..

[Tip for an effective data collection]

- Keep a fieldwork journal that you will use to take notes during your site visits/group discussions/class.
- This will help you track changes in your observations and analysis as you (re)visit the site and progress throughout the semester.
- You can write/record anything on this journal – your observations, opinions, questions, drawings, sketches etc..

Here are some points to consider before, during and after your site visits. While the processes are presented as linear, they may also occur reiteratively and at multiple stages of the site visit/research process.

During Site Visit

Engage in different rounds of observations and take detailed notes. If possible, do take pictures/videos of key areas and issues. As you engage in observations, also record down questions that may arise for you to address later.

1. Initial Observations

- Note the time of your visit. This matters as when you visit the area that may significantly affect your findings.
- Note down your first impressions and overall atmosphere of the place.
- Observe the physical layout of the place, e.g. buildings, streets, parks, infrastructure, human traffic and demographics, availability of sheltered walkway.
- Also record which demographic is there (e.g. who is there, what are the age groups, ethnicity, possible occupations etc..).

Here are some points to consider before, during and after your site visits. While the processes are presented as linear, they may also occur reiteratively and at multiple stages of the site visit/research process.

During Site Visit

2. Detailed observation of hardware/spaces

Here is a list of **possible** aspects that you may want to look out for:

- **Infrastructure** e.g. sidewalks, sheltered walkways, graffiti/street art, dark alleys, poorly lit areas, ramps/handrails
- **Housing** e.g. types of housing, housing conditions, occupancy rates,
- **Public spaces** e.g. availability and conditions of parks, recreational areas, cleanliness, usage
- **Community areas** e.g. community centers, polyclinics, libraries
- **Commercial areas** e.g. types of businesses, foot traffic, economic activity
- **Traffic and transportation** e.g. flow of traffic, congestion points, availability and affordability of transport, peak times for traffic, busy vs quiet areas, unsafe crossings
- **Signage and Wayfinding** e.g. clarity and effectiveness of signs and directions
- **Environmental aspects** e.g. green spaces, pollution levels, waste management, presence of litter, noise and air pollution, air quality

In looking at these specific domains, ask yourself the 5W1H questions:

- What is happening?
- Where is this happening?
- When is this happening?
- Why is this happening?
- Who is there, and who is this happening to?
- How is this happening?

Here are some points to consider before, during and after your site visits. While the processes are presented as linear, they may also occur reiteratively and at multiple stages of the site visit/research process.

During Site Visit : Some Tips

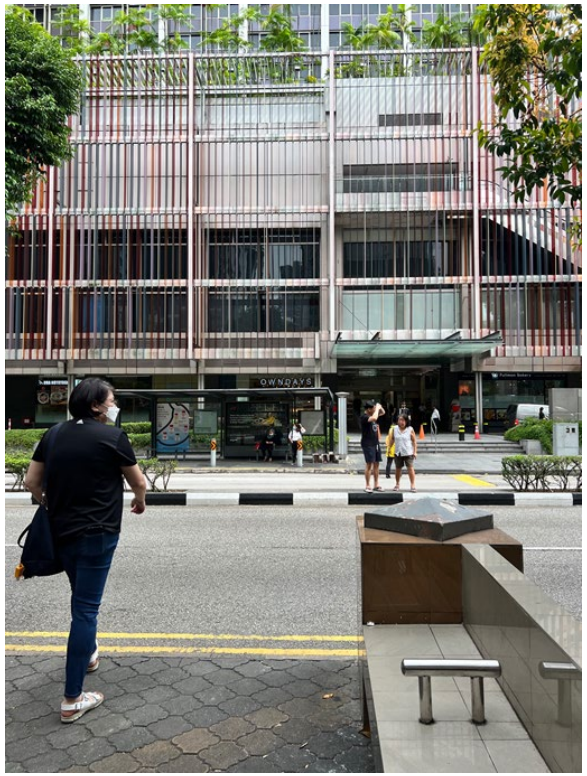
What should we look out for?

Use the 5 Ws Tool:

- **Who?** Stakeholders
- **What is happening?** Behavior/activities of people in the area and the problems that they face
- **Where?** Provide context – What you see in your living environment impacts how you read/understand the urban context
- **When?** Time matters, when and how long you visit the site for will give you different observations; e.g. CBD on a weekday lunch period VS CBD on a weekend lunch period
- **Why?** What are the reasons for the behavior/activities of the people and what are the causes of the problems that they face?

During Site Visit | An Example

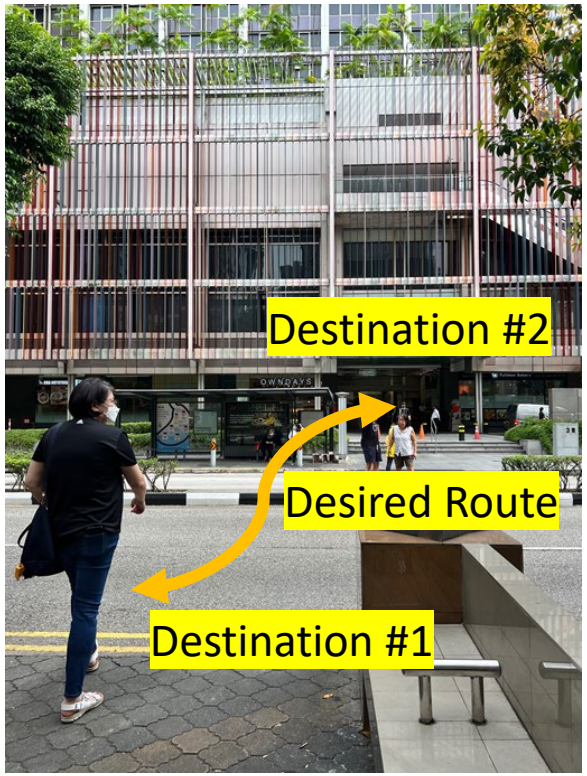
Apply the questions to this street in Tanjong Pagar:

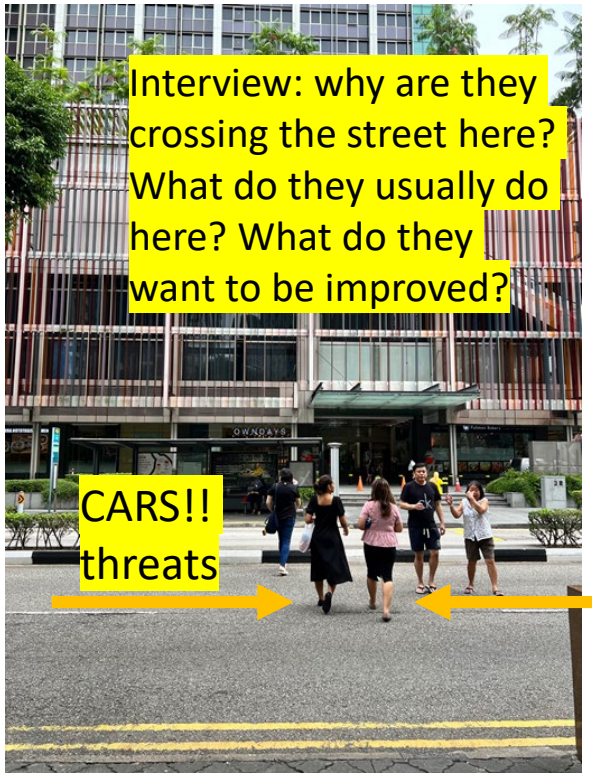


- Who?
- What is happening?
- Where?
- When?
- Why?

During Site Visit | An Example

Apply the questions to this street in Tanjong Pagar:





- **Who?** Officer workers? Tourists?
- **What is happening?** People going for lunch? People going to shop?
- **Where?** Is there something about the location of this crossing that makes it a popular one?
- **When?** When do people normally use this crossing frequently? Is it during lunch hour? Or closer to the evening?
- **Why?** Why do these people (Who) do what they do (What are they doing) at this particular crossing (Where) at this time (When) despite the dangers of cars? Why is there currently a lack of safeguards in this area for people crossing despite it being frequently used?

During Site Visit

3. Detailed observations of software/human interactions/social relations

- Observe interactions among people e.g. their behaviors and activities, what they are doing.
- Note demographic details: age group, cultural diversity etc..
- Pay attention to accessibility issues: are there facilities for older people and persons with disabilities?
- If possible, conduct interviews/surveys with the community members (refer to Research Methods Guide for details).
    - Always ask for consent before proceeding with the interview/survey.
    - Ask open-ended questions.
    - Phrase your questions in a neutral way.

Here are some points to consider before, during and after your site visits. While the processes are presented as linear, they may also occur reiteratively and at multiple stages of the site visit/research process.

After Site Visit

1. Review and reflect

- Review your notes and reflect (make sense of the collected data) on your observations and findings: what possible problems did you identify?
- What questions did you have when you were on the site visit? How can they contribute to the formulation of your project?

2. Analyze

- Categorize the identified problems into key categories or themes.
- Identify patterns and connections between different categories and/or themes.
- Identify linkages between the data collected or findings from the site visit to the problem statement(s) you have identified.

3. Summarize

- Summarize your key findings and corroborate with evidence (pictures, videos, sketches)
-

Here are some points to consider before, during and after your site visits. While the processes are presented as linear, they may also occur reiteratively and at multiple stages of the site visit/research process.

Some Strengths and Weaknesses of Site Visits

Strengths of Site Visit	Weaknesses of Site Visit
Direct observation: Allows for firsthand experience and real-time data collection	**Subject to observer bias** (Vogt et al, 2012): Personal biases of the researcher can influence observations and interpretations To mitigate this, reflect on your own blind spots and assumptions before heading down for your site visit.
Contextual understanding: Provides a deeper understanding of the physical and social context of the area	**Time consuming**: Site visits can require significant time, especially if the team has not decided beforehand where they want to visit, or if the area is too large
Interaction with community: Enables direct communication with the community (e.g. residents, business owners, bus drivers etc..)	**Limited generalizability**: Findings from a specific site may not be generalizable to other contexts or locations
Sensory experience: Allows for engagement of all senses, offering a more comprehensive reading of the area	
Verification of secondary data: Allows for cross-checking of data obtained from desktop research	
Identification of issues not evident in secondary data: You may encounter problems/opportunities that are not recorded in documents/reports/news articles	

You may complement your site visits with other research methods such as desktop research and interviews (refer to Research Methods Guide) for a more comprehensive and nuanced understanding of your project site!

References

- Cohen, L., Manion, L., & Morrison, K. (2018). *Research methods in education* (8th ed.). Routledge.
- Musante, K., & DeWalt, B. R. (2010). *Participant observation: A guide for fieldworkers*. Rowman Altamira.
- Vogt, W. P., Gardner, D., & Haeffele, L. (2012). *When to use what research design*. Guilford Press.

Research Methods

This guide has been adapted and simplified for CDE2501.

Research Methods | Desktop Research

Purpose:

- Essential before your first visit and before undertaking other data collection methods.
- To get a general overview of the district that you will be working on.
- Will provide you with a clearer understanding of the history, demographic profile, unique features and potential issues/challenges faced by different actors in the district (amongst other information).
- Enables a narrowing down of the scope of research.

Sources:

- Involves combing through publicly available and online sources, such as documents, reports, news articles, commentaries and opinion pieces.
- Starting points: news outlets and academic databases, through search engines like Google, Google Scholar, and the NUS Library Portal.
- Other sources: grey literature, such as white papers and government documents.
- Good way to source for both quantitative and qualitative data
    - Quantitative data: population numbers, population density, labor force participation e.g. Department of Statistics website
    - Qualitative data: public sentiments/opinions, future plans, past maps, Parliamentary speeches, and budget releases

Research Methods | Desktop Research

Evaluating sources:

- Be mindful of the reliability and credibility of your sources.
- Note possible biases of each source.
  - e.g. The Straits Times is Singapore's national newspaper, authors may articulate particular narratives or ideas
- Some questions to ask yourself:
  - Who is the author?
  - What is the source? Is this a reputable news outlet or an academic publisher? Are they well-known and respected?
  - Is the source current? When was the information published or last updated? Is the information still relevant today?
  - Who is the intended audience?
  - Is the information supported by evidence? Are the claims corroborated by data and/or evidence?
  - Does the source show any signs of bias or a particular agenda? Is the information presented in a balanced and impartial manner?
  - Are there any factual errors and inconsistencies?
  - Are there any limitations or gaps? Does the source acknowledge any limitations or potential biases? Are there important aspects of the topic that are not addressed?

Tip:
Triangulate your sources – check different sources of information to confirm that your sources and understanding of the issue are accurate.

Research Methods | Desktop Research

Organizing information:

- Desktop research will reveal extensive information about your district/potential topics, and it is thus important to organize the information in an accessible and organized manner.
- Depends on you and the nature of the topic, findings can be organized in two ways:
  - Thematic, or
  - Chronological
- Consider using free citation management tools like EndNote and Zotero to manage references.

Analysing and synthesizing information:

- After finding common recurring theme/issues, synthesize all data collected from different sources to form a coherent argument.
- You may use findings from desktop research to corroborate your claims and arguments.

Research Methods | Content Analysis

Purpose: To understand the value of analyzing textual, visual, or audio content to identify patterns, themes, and meanings related to urban planning issues.

Preparing for Content Analysis:

1. Define your objectives: Clearly outline what you want to achieve from undertaking content analysis, and the information that you are seeking for.

2. Formulate questions: Develop specific questions that will guide your content analysis.

3. Select your content: Identify and gather relevant content sources such as newspapers, magazines, Parliamentary speeches, National Day Rally speeches, social media posts, reports, advertisements, photographs, and videos.

4. Define and clarify context of your content:
    a. The source of your content
    b. The originally intended purpose, setting, and audience of the content

Research Methods | Content Analysis

Develop a coding framework and ethical considerations:

1. Create categories: Develop a coding scheme that include categories and sub-categories. These could be themes, concepts, keywords, or issues.
2. Remain adaptable: Be flexible and willing to adjust your coding scheme if new themes or patterns emerge during the analysis.
3. Define your unit of analysis: You may have multiple units of analysis, e.g. words, phrases, sentences, paragraphs, or the entire text (Tunison, 2023).
4. Respect copyright: Ensure that you have permission to use the content.
5. Protect confidentiality: If analyzing private content, ensure the confidentiality of the source of information.
6. Maintain objectivity: Avoid introducing personal biases into the analysis. Aim for objectivity.

Conduct the Content Analysis:

1. Select a sample: Choose a representative sample size that is sufficient to draw insightful and generalizable conclusions.
2. Code your content. Apply your coding scheme to the sample. This will involve systematically categorizing content based on predefined codes. You may consider using Excel and Word to organize the data collected.
    a. The categories may overlap with one another, and some codes may appear in more than one category (Tunison, 2023).

Research Methods | Content Analysis

Analysing the data collected:

1. Identify patterns and themes: Look for recurring patterns, themes, and trends within the coded data. Additionally, be mindful of inconsistencies, contradictions and differences between/among sources or ideas (Tunison, 2023). This could involve both quantitative (e.g. frequency counts) and qualitative (thematic) analysis.

2. Interpret findings: Relate your findings to your questions and objectives. Discuss what the identified patterns and themes reveal about your topic/project.

3. Summarize key points: Highlight the most important findings from the content analysis. You may use visualization tools such as charts, graphs and tables to present the data collected clearly and effectively.

Additional online resources:

- Adu, P. (2016). Qualitative analysis: Coding and categorizing data. https://youtu.be/v_mg7OBpb2Y
- Duke University—Mod-U (2016). How to know you are coding correctly: Qualitative research methods. https://youtu.be/iL7Ww5kpnIM
- Gramenz, G. (2014). How to code a document and create themes. https://youtu.be/sHv3RzKWNcQ Shaw, A. (2019). NVivo 12 and thematic/content analysis. https://youtu.be/5s9-rg1ygWs

Research Methods | Interviews

Purpose: The purpose of interviews is to understand public perceptions of your allocated district and/or chosen topic, to complement findings from desktop research.

Preparing for the interview:

1. Define your objectives:

- Clearly outline the goals of the interview and the information that you are seeking
- It is okay to have a broader vision/objective in the initial stages of formulating your project.
    - Questions like "What do I want to take away from this interview" and "How will responses from the interview contribute to my project" may help with crafting interview questions.
- Additionally, based on the findings from your desktop research, you may also want to consider the following questions:
    - Where and when will the interviews be conducted?
    - Who will be interviewed, and who will be conducting the interview?
    - How many people do you plan to interview (i.e. what is your sample size), and why?
    - Who is the profile of your target group/intended respondents (e.g. age, class, ethnicity, health needs), and why?

Research Methods | Interviews

2. Develop your interview questions:

- Create a list of open-ended questions tailored to your intended respondents.
- If you are planning to interview different groups (e.g. bus drivers and commuters), prepare separate questions.
- Some questions to consider when crafting your interview questions:
    - Craft topic-specific/focused questions: Instead of asking “Tell me about the district”, ask “What are some areas for improvement for the public transport infrastructure of the district?”
    - Avoid asking leading questions: Instead of asking “Why is the new bus route better than the previous one?”, you may rephrase it to “How do you feel about the changes made to the bus route?”
    - Encourage detailed responses: Add in prompts into your question. If you are looking at public transport infrastructure, instead of just asking about the areas for improvement, ask/follow-up with a “why”, or “what do you mean by ...?”.
    - Be neutral and non-judgmental: Instead of asking “Why don’t you use the public transportation system?”, ask “How would you describe your experience with using or choosing not to use public transportation?”

It is essential that your interview questions are carefully designed before fieldwork is conducted, if not you risk wasting your respondents’ and your own time.

Research Methods | Interviews

3. Conducting the interview: Ethical considerations

- Ethical considerations: Inform respondents about the purpose of the survey, how their data will be used, and their right to withdraw.
- Anonymity and confidentiality: Ensure respondents that their responses are anonymous and confidential.
- Sensitive questions: Handle sensitive questions with care, providing options for respondents to skip questions if they feel uncomfortable.

3. Conducting the interview: Data collection

- Approach respondents in a respectful manner and in an appropriate location. Always introduce yourself and explain why you are talking to them, and only proceed with the interview after receiving consent. Do mention that their identity will be protected, and that confidentiality will be maintained. Inform respondents about the purpose of the interview, how their data will be used, and their right to withdraw.
    - If you want to audio record/take pictures/videos of the interview, always ask for permission. Explain why you want to record, and how the recording will be used.
- Build rapport: Ask them about their backgrounds if it is relevant to your project (e.g. age, purpose of being in the area etc.).
- Use your prepared questions as a guide, but be flexible to follow interesting leads.
- Pay close attention, and ask relevant follow-up questions.
- Take notes: Jot down key points, quotes and observations during the interview.
- Once the interview has concluded, thank the respondents for their time.

Research Methods | Interviews

4. After the interview: Analysing and presenting data

- After conducting multiple interviews, analyse your interview data.
- Identify common themes: Look out for commonly mentioned ideas, challenges, pain points, aspirations etc..
- Highlight the most important findings from the interview process. Do take note of contradictory or inconsistent findings as well.
- Quote your interviews: In your presentation, use direct quotes to support your analysis and provide authenticity.

Research Methods | Surveys

Purpose: Surveys can be conducted to:

- Gather public opinions/feedback and assess public satisfaction and needs
  - Collect insights from residents, businesses, commuters, and other relevant actors within the community.
- Identify urban problems
  - Feedback gathered from the community can help identify urban issues, concerns and priorities.
- Support decision-making
  - Survey results can provide quantitative data to corroborate your project.
- Engage the community
  - Surveys can promote community engagement and participation.
  - Surveys may therefore empower the community by giving them a voice in shaping the future of their neighborhood/districts.

Research Methods | Surveys

Preparing for the survey: Based on initial findings from desktop research,

- Define your objectives
  - What are the goals of your survey, and what information are you seeking to gather from the surveys?

- Identify your target audience, and method of distribution/dissemination
  - Who do you want to survey, and why?
  - What is your sample size, and profile of your intended audience, and why?
  - If you want to survey different actors, prepare separate survey questionnaires.
  - Think about how the survey will be conducted? If in-person, where and why? If online, how will you ensure that your survey reaches your intended audience?

- Develop your survey questions
  - Questions should be clear, concise, unbiased, and possible free of jargon
  - This would help to reduce response errors, whereby the questions asked may not be understood in the way intended, or the respondents may feel pressured into agreeing with your own ideas (Parfitt, 2005).
  - Avoid **double-barreled questions** e.g. “Have you travelled by train or bus within the last 1 week?” There are two questions in this statement, and they should be asked separately.
  - **Avoid negatives and double negatives** e.g. “Would you rather not use non-nuclear sources of electricity?”

Research Methods | Surveys

- Avoid **leading questions**
  - Leading questions denote your personal opinions and are likely to induce a biased response
  - Instead of asking "Why is the new bus route better than the previous one?", you may rephrase it to "How do you feel about the changes made to the bus route?"
- Be concise in your phrasing
  - Terms like 'usually' or 'recently' are vague. Specify this by providing a time frame such as 'in the last week/month'
- Be concise in your phrasing
  - Terms like 'usually' or 'recently' are vague. Specify this by providing a time frame such as 'in the last week/month'

- Think about the types of questions you want to ask (e.g. multiple choice, Likert scale, open vs closed-ended), depending on the information that you are seeking.
  - Three main types of data (Parfitt, 2005)
    1. **Data that classify people, their situations and their environment**: Also known as respondent variables, and collect information about respondents' age, income group, household type, health status etc.
    2. **Data that relate to people's behavior**: Collect information about e.g. where to they go to shop, do they visit the mall, how do they travel there etc.
    3. **Data that relate to people's attitudes, opinions and beliefs**: Collect information about e.g. their thoughts on the cost and reliability public transport system, their experiences on the Singapore Flyer
  - Organize questions logically, starting with general questions and moving on to more specific ones.
  - Keep the survey as short as possible to avoid respondent fatigue. There is no hard and fast rule as to how many questions should be asked as this will depend on the nature of your project and the information that you are seeking.

Research Methods | Surveys

Ethical considerations

- Ethical considerations: Inform respondents about the purpose of the survey, how their data will be used, and their right to withdraw.
- Anonymity and confidentiality: Ensure respondents that their responses are anonymous and confidential.
- Sensitive questions: Handle sensitive questions with care, providing options for respondents to skip questions if they feel uncomfortable.

Distributing the survey

- Choose a distribution method: In-person at site or through online platforms (e.g. Google Forms, SurveyMonkey, Qualtrics).
- Targeted outreach: Reach out to specific groups relevant to your research objectives. One way to start is snowball sampling, using your current social networks.

Collecting responses

- Monitor the responses: Track the number of responses and ensure that you are reaching your target sample size.
- If you are encountering low response rates, you may consider adjusting your distribution strategy, or using two distribution methods.

It is essential that your survey is carefully designed before fieldwork is conducted, if not you risk wasting your respondents' and your own time.

Research Methods | Surveys

Analysing survey data

- Data cleaning: Check for incomplete or inconsistent responses and decide as a group on how to handle them.
- For quantitative data, analyse numerical data (e.g. mean, median, total etc.) and identify patterns and trends.
- For qualitative data, analyse open-ended responses and look for common themes and insights.

Summarizing the collected data

- Summarize key points: Highlight the most important findings.
- Visualize your data: Use charts, graphs and tables to present data clearly and effectively.
- Discuss implications: Explain how the survey results inform your project.

Research Methods | When to conduct Surveys vs Interviews

- Similar ways to prepare for both research methods.
- Major differences are in the ways of collecting information, the type of data collected, and the analysis of data.
- Which method you choose to use depends on the nature of your project:
  - **Research objectives**: Determine whether your research aims to gather quantitative (survey) or qualitative insights (interview). Consider the specific questions guiding your study.
  - **Sample size**: Surveys are more suited for large sample sizes and broader data collection across bigger and diverse groups. Interviews are more appropriate for smaller, targeted samples or when depth of understanding is crucial.
  - **Type of information needed**: If you need to measure attitudes, behaviors or preferences numerically, a survey is appropriate. For exploring personal experiences and opinions, interviews may provide deeper insights.
  - **Resource and time constraints**: Consider the resources, time and logistical constraints involved in conducting surveys versus interviews. Surveys are generally more cost-effective and efficient for large-scale data collection, while interviews require more time and effort per respondent.
  - **Complementary approaches**: In some cases, using both surveys and interviews in a mixed-methods approach can provide a more comprehensive understanding of the research project, combining the strengths of quantitative and qualitative data.

References

- Parfitt, J. (2005). Questionnaire design and sampling. In R. Flowerdew & D. M. Martin (Eds.), Methods in human geography (pp. 78-109). Routledge.
- Tunison, S. (2023). Observational study. In J. M. Okoko, S. Tunison, & K. D. Walker (Eds.), Varieties of qualitative research methods (pp. 45-67). Springer.

Data Collection, Analysis, and Visualization

*CITATION:

- Always cite your sources – primary, secondary, data sources, photos, videos, diagrams, quotes, qualitative data sources, facts, etc.. (*Allow others to verify the credibility of your sources and avoid plagiarism)*
- Use APA format: https://apastyle.apa.org/style-grammar-guidelines/references

Why do we need data?

- Informed Decision-Making
  - Good data provides evidence and can help to justify decisions
  - Bad data, anecdotal evidence, assumptions or abstract observation lead to bad decisions
- Problem-Solving
  - Understand the root cause of a problem
  - Gain hindsight
  - Predict future events
- Back Up Your Arguments
  - Allows you to substantiate your claims or arguments
  - Helps to make a case for why you want to take specific actions
- Keep track of your progress
  - Are you getting to where you want to be?
  - How much time, money or effort do you need to add?
  - Do you need to rethink your plan?

Data, Information, Knowledge, Wisdom (DIKW) Pyramid

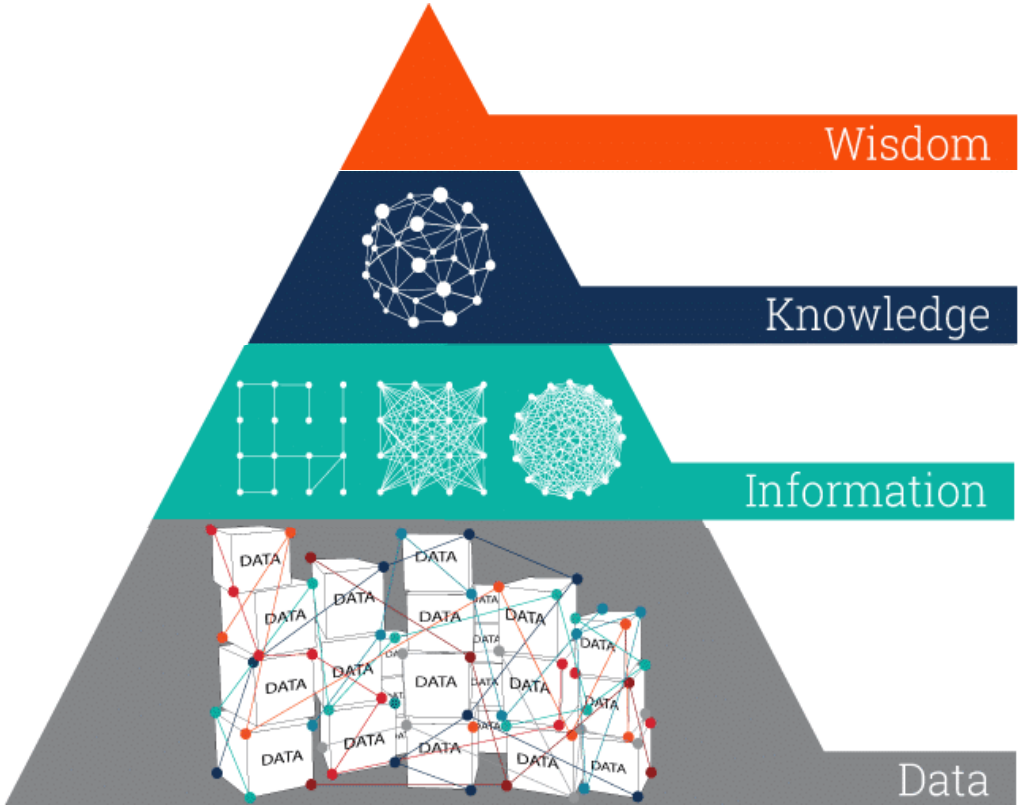


Image source: https://www.ontotext.com/knowledgehub/fundamentals/dikw-pyramid/

Wisdom

"Rightful" Application of Knowledge

Knowledge

Extracting New Understandings

Information

Giving it context

Data

Each step up the pyramid answers questions about and adds value to the initial data.

- Uncovering relationships that are not explicitly stated as information.
- Understanding how to apply the information

- Cleaned and structured data
- Easier to measure, visualize and analyse for a specific purpose
- By asking questions using the 5 "W"s, valuable and useful information can be obtained.

- Raw and unorganised (e.g.: numbers, characters etc.)

Data Analysis Process

Problem Statement

Site surveys/ Desktop research

Turn data into information

Turn information into knowledge

Data Visualization

Identify & Define the Question

What problem are you trying to solve? Define your objectives and why you want to archive them

Collecting the data

Identify what data you need, how much, and your data collection strategy

Cleaning the data

Prepare the data for analysis by checking for errors and structuring your data

Analysing the data

Type of data analysis performed largely depends on what your goal and data type and format

Interpreting the Results

One of the hardest parts is to make sense of your data and deciding what actions need to be taken based on the results.

Sharing your results

Presenting your results in a way that is easily understood and interpreted by others. Data visualisation is helpful to communicate findings

Research Methods (How do we get data?)

Data & Literary Research – Secondary Sources of Data:

- Archival Research
  - helps contextualize the district
- Discourse Analysis
  - analyzing language or behavior (written, vocal, visual, non-verbal) of events or social life
  - perspective of the narrator/author/participant
- Data Analysis
  - use open-data sources, create models, run regressions, use data as projective or to support your claim
- Case Studies
  - similar references to identify strengths and weaknesses

Example of Websites with open data sources (secondary data):

- Singapore Census of Population 2020: https://www.singstat.gov.sg/publications/reference/cop2020/cop2020-sr1
- Business-related data from ACRA: https://www.acra.gov.sg/training-and-resources/facts-and-figures
- Public data from Singapore: https://data.gov.sg/
- LTA DataMall (may need basic Python): https://datamall.lta.gov.sg/content/datamall/en.html
- Population Density: https://www.citypopulation.de/

Analysing Data

Types of Data Analysis Methods

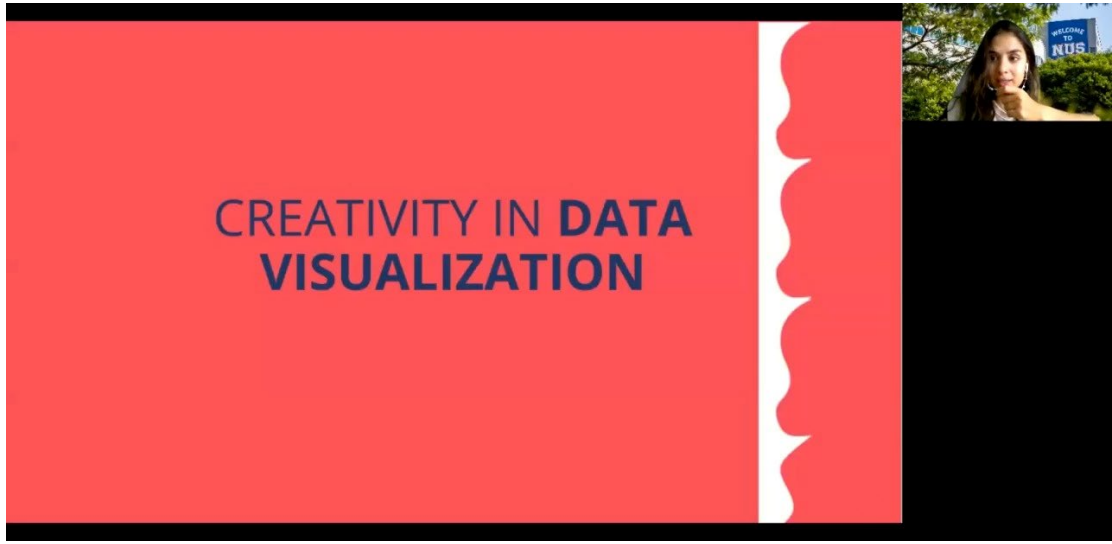
Qualitative or Categorial Data	
Narrative	Interpreting stories or experiences
Thematic	Identifies patterns and themes
Ethnography	Examines cultural phenomena through immersive fieldwork and observation
Case study	Using a single/multiple case studies to gain insights
Content	Examines the content of written or visual material to identify patterns and themes
Conversation	Examines social interaction, focusing on verbal and non-verbal conduct.
Visual	Examines visual materials like images, videos, and graphs

Quantitative or Numerical Data	
Statistical (Basic)	Discover underlying patterns and trends • Descriptive statistics – e.g.: mean, median, standard deviation • Regression • Correlation • Comparison
Cluster	Explores naturally occurring groups within a data set
Factor	Describe variability among observed, correlated variables
Network	Studying the relationships between entities inn a network.

**Listed data analysis methods are non-exhaustive; these are some possible methods that you can use in your studio project.*

Telling story with data
Data Visualization

- Effectively communicate information and insights through graphical representations
- Make complex data easily understandable
- Identify patterns and trends
- Communicate insights and findings
- Support decision making
- Explore data



Video Sharing on Data Visualization by TA Valerie can be found on Canvas!
Watch the video to find out various ways to visualize your data!

Important Considerations for Data Visualization

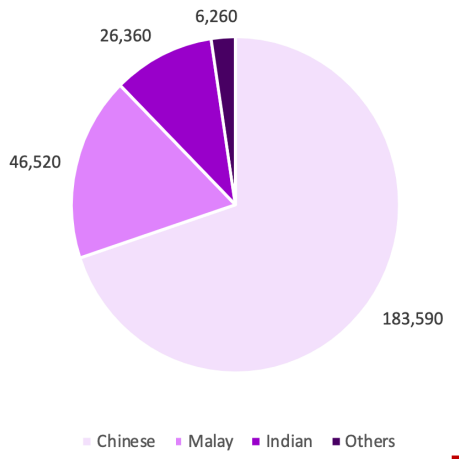
How will you represent your ideas in a clear and intuitive way?

1. **Purpose:** Understand what the data represents and what message needs to be conveyed.
2. **Data quality**: Ensure data is accurate, complete and relevant.
3. **Audience**: Consider who the audience is and tailor the visualization accordingly.
4. **Design**: Choose appropriate visual encoding, colors, labels, scales, etc..
5. **Simplicity**: Avoid clutter and keep the visualization as simple as possible.
6. **Legibility**: Make sure the visualization is easy to read and interpret.
7. **Interactivity**: Provide ways for the audience to interact and explore the data.
8. **Context**: Provide enough context for the data to be understood (units, time periods).
9. **Ethics**: Consider ethical implications, such as privacy and bias in the data and visualization

Common Types of Data Visualization

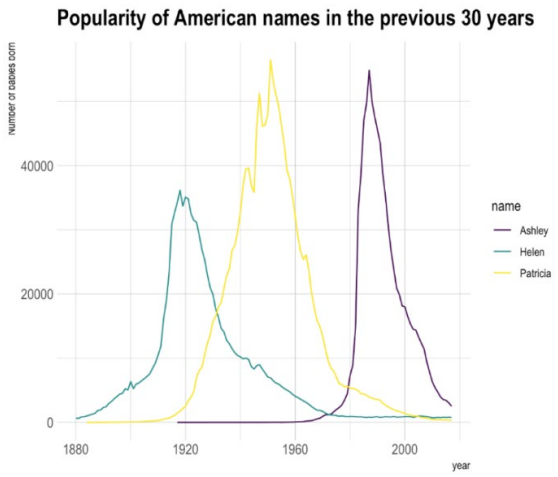
Hint: there are charts and maps for you to use in the studio guide!

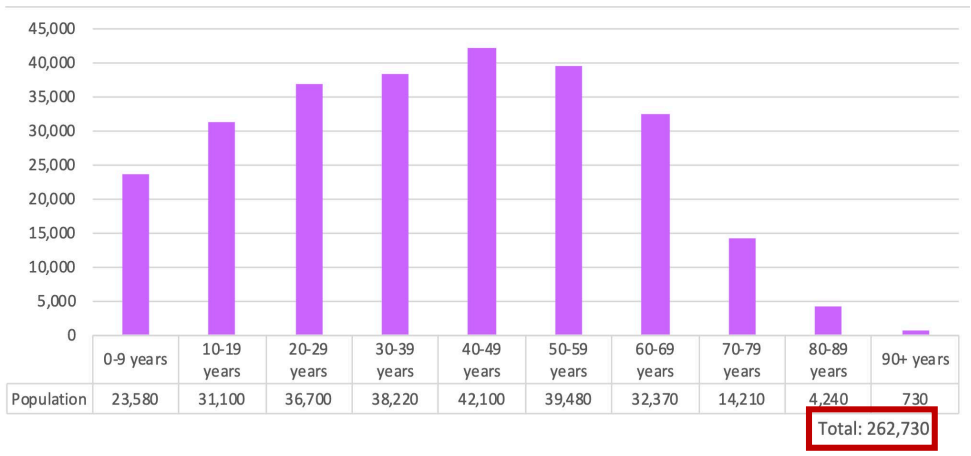
Pie, Line and Bar Charts: organizes data in a simple way to compare the size and scale of different categories; and to show trends over time



Ethnic Group	Chinese	Malay	Indian	Others	Total
Population	183,590	46,520	26,360	6,260	262,730

Total Ethnic Group Distribution for the Jurong West District based on the map boundaries.
Source: Singapore Census of Population 2020, Department of Statistics





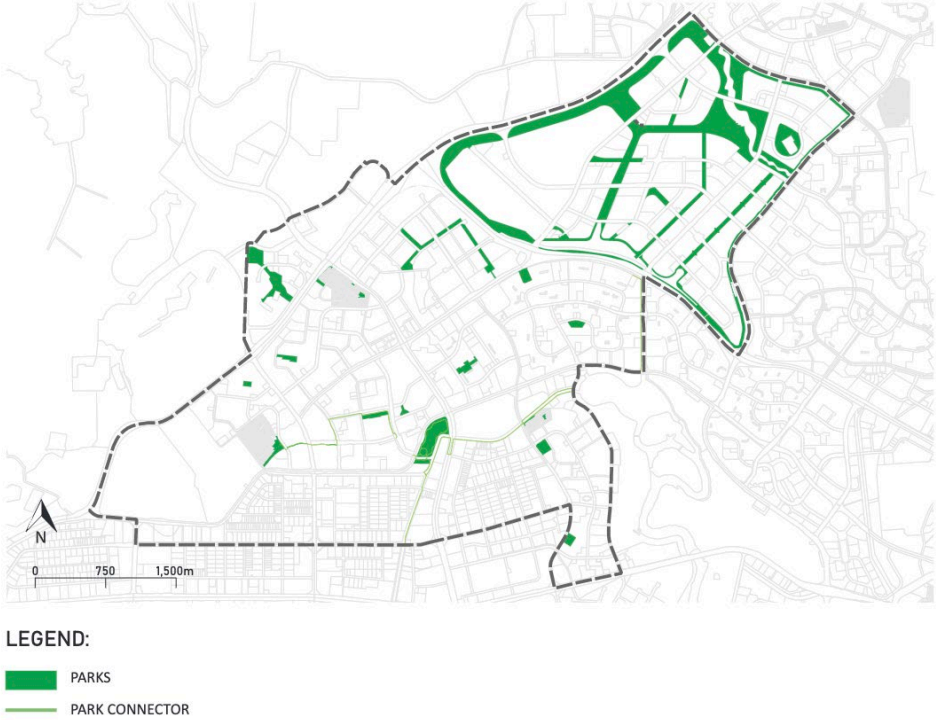
Total Age Group Distribution for the Jurong West District based on the map boundaries.
Source: Singapore Census of Population 2020, Department of Statistics

Make clear what data your figures represent
(e.g. Source, method of collection, sample size).

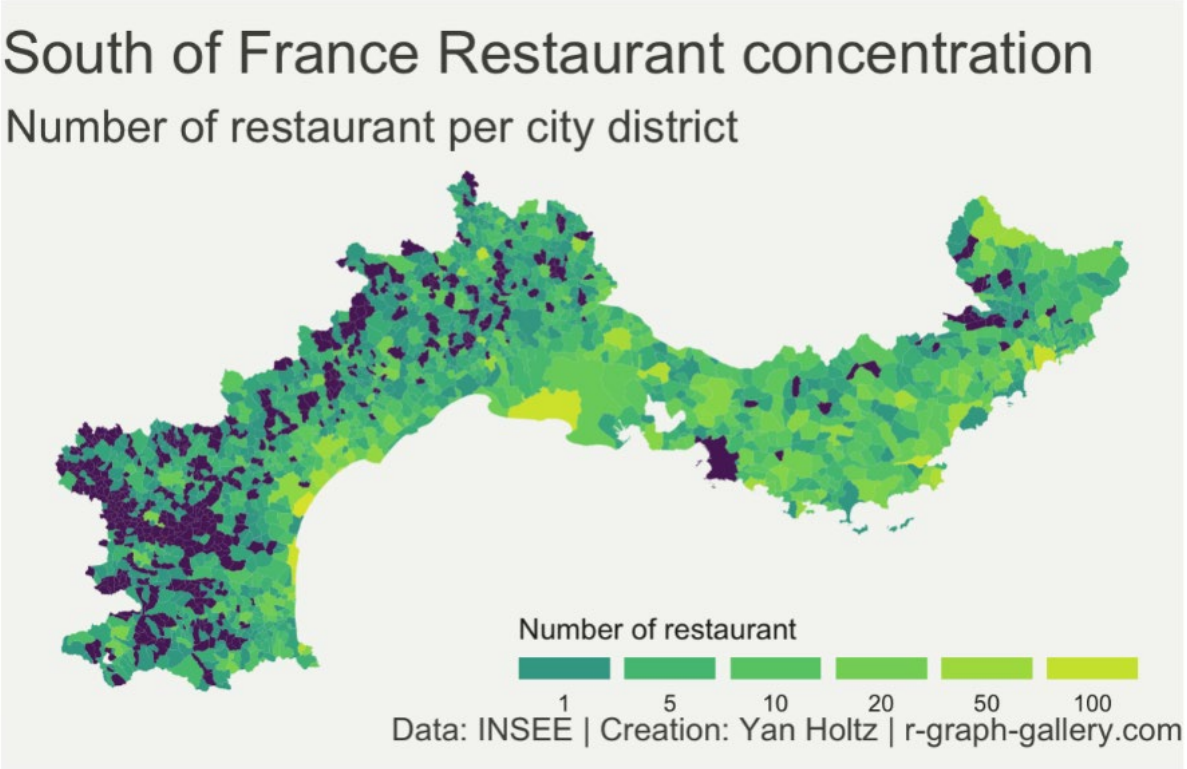
Common Types of Data Visualization

Hint: there are charts and maps for you to use in the studio guide!

Maps: organizes data spatially; helpful in visualizing data by location and scale, presents spatial patterns

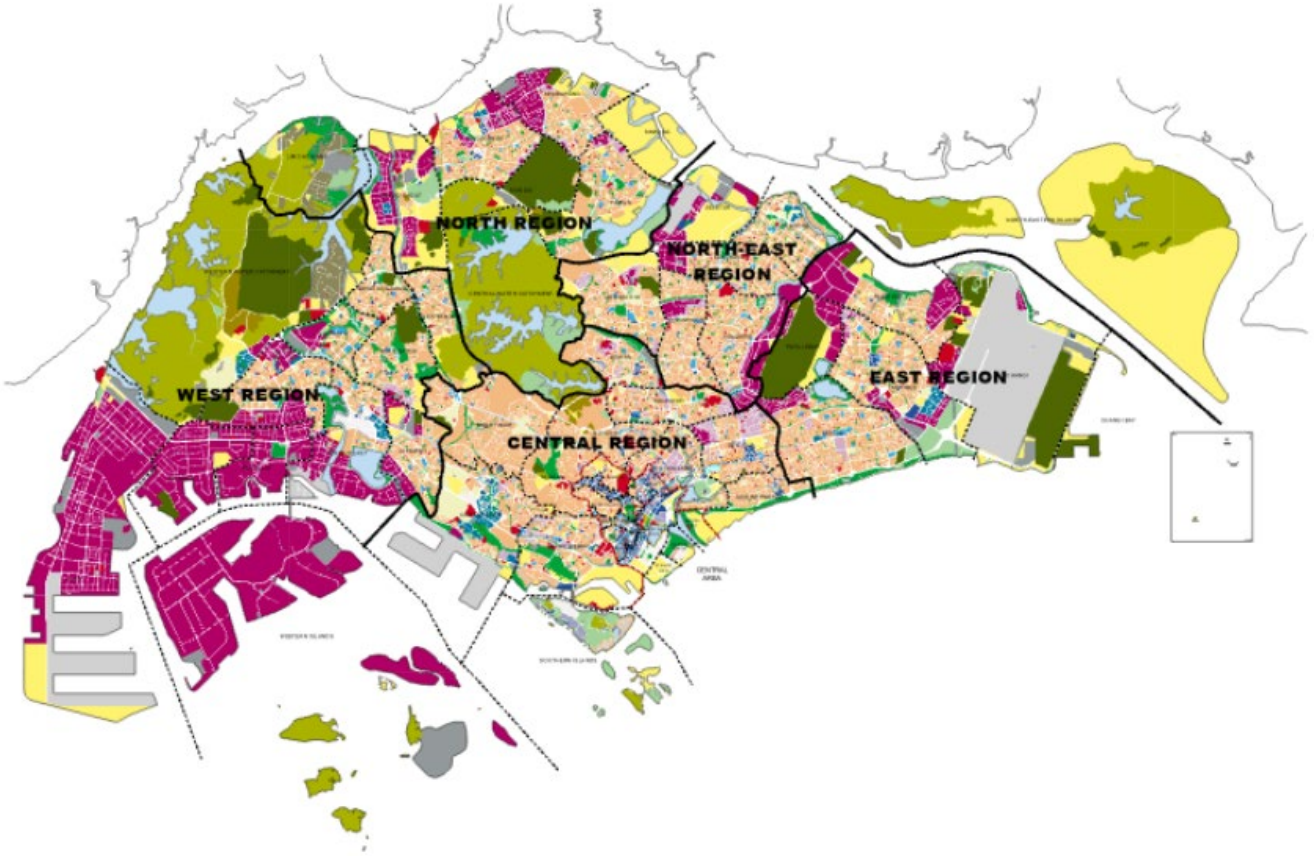


Sources: *Park Connector Loop* accessed on 16 May 2023 from *National Parks Board* from https://data.gov.sg
Master Plan 2019 Land Use Layer accessed on 16 May 2023 from *Urban Redevelopment Authority* from https://data.gov.sg



Common Types of Data Visualization

Land Use Plans: provides information and context on land use and planning intention to supplement other data



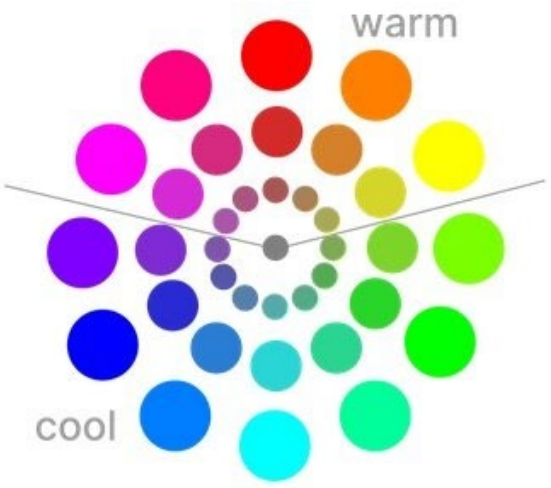
Legend

- Residential
- Residential with Commercial at 1st storey
- Commercial & Residential
- Commercial
- Hotel
- White
- Business 1
- W Business 1 - White
- Business 2
- W Business 2 - White
- Business Park
- W Business Park - White
- Residential / Institution
- Commercial / Institution
- H Health & Medical Care
- E Educational Institution
- W Place of Worship
- Civic & Community Institution
- Open Space
- Park
- Beach Area
- Sports & Recreation
- Waterbody
- Road
- Transport Facilities
- Railway
- Mass Rapid Transit

Singapore's Master Plan https://www.ura.gov.sg/maps

Telling a story with colors

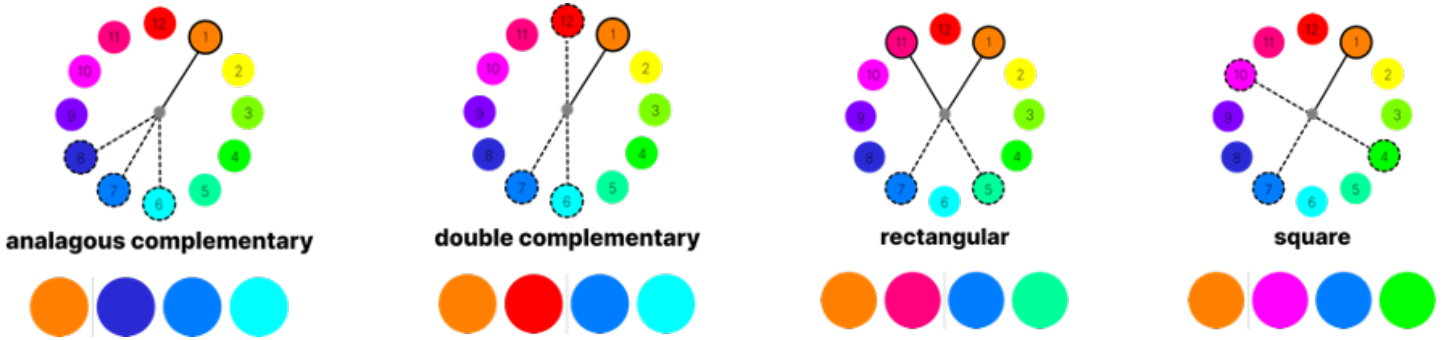
We can leverage on color temperature or saturation to help imply relationships and differences in our data.



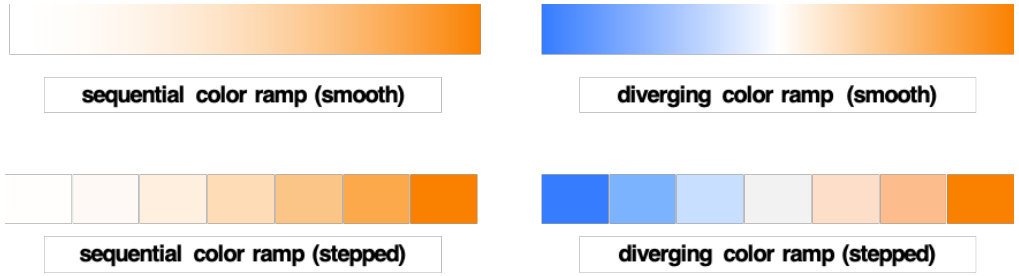
color temperature

Source: www.storytellingwithdata.com

Different combination of colors tell a different story when comparing 4 different subjects.



Use changes in saturation or lightness to show changes in quality.



Read more at: https://www.storytellingwithdata.com/blog/2020/5/6/picking-the-right-colors
Explore different types of data visualization styles at: https://informationisbeautiful.net/visualizations/

Data Visualization Tools & Resources

There are many tools and resources available, these are just a few:

TOOLS

1. Microsoft Excel / Google Sheets
2. PowerPoint / Google Slides
3. Canva
4. Adobe Spark
5. GIS Spatial Data
    1. QGIS / ArcGIS
6. Adobe Photoshop / Illustrator
7. 3D Modelling
    1. Sketchup / Rhino

RESOURCES

1. Own data collection (surveys, interviews etc..)
2. Local Data Sources:
3. SG National open data collection https://data.gov.sg
    1. Land Transport Data Mall https://datamall.lta.gov.sg
    2. URA Space https://www.ura.gov.sg/maps
4. International Data Sources:
    1. Open Street Map / Google Map / Google Earth
    2. World bank https://data.worldbank.org/
    3. UN https://data.un.org/

Visualization for Proposal Concepts

How can we enable our audience to imagine what our solutions look and feel like?

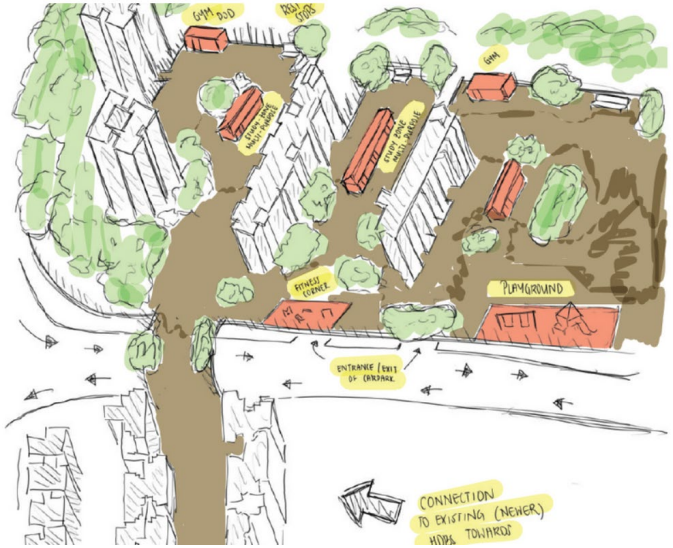
Visualization for Proposal Concepts

How do we also present and communicate our proposals and ideas?

Here we will explore diverse means of visualizing ideas in the most effective way!

Common Types of Visualizations:

- Photo Montage / Collages
- Sketches
- Plans
- Mood Boards
- 2D/3D modelling (optional)



Source: East Coast District 4 Team 4 AY2022/23 Sem 1

Source: https://www.archdaily.com/874177/soi-16-studio-plp/594a5bdeb22e38e9290000e8-soi-16-studio-plp-concept

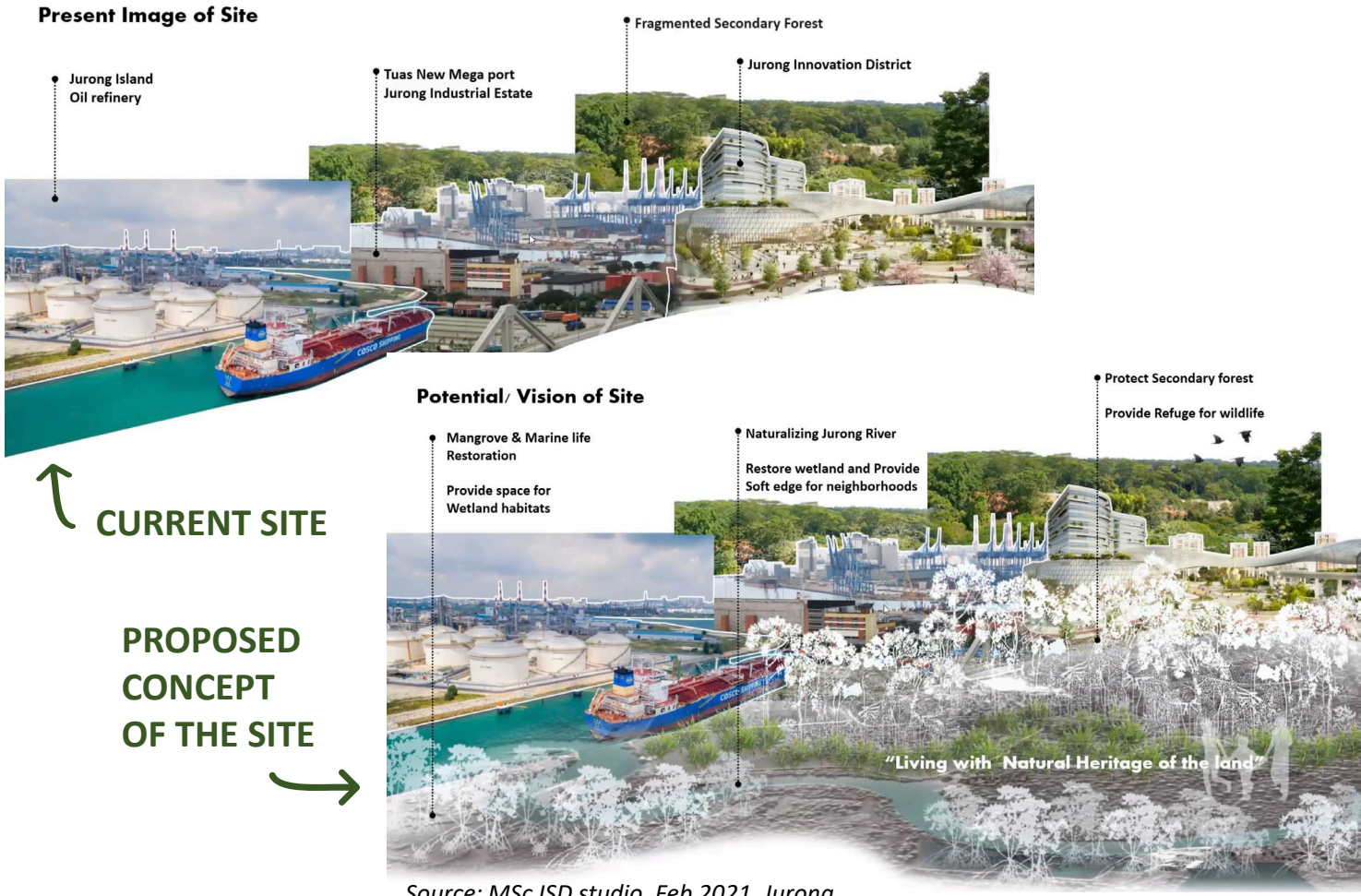
Visualization for Proposal Concepts

PHOTO MONTAGE / COLLAGE

This is a fun way to imagine new spaces and represent its conceptual elements!

This is an example of overlaying photos, **words**, sketches, or objects, to create an assemblage that contributes to a cohesive vision.

**Remember not to overcomplicate your montage!*

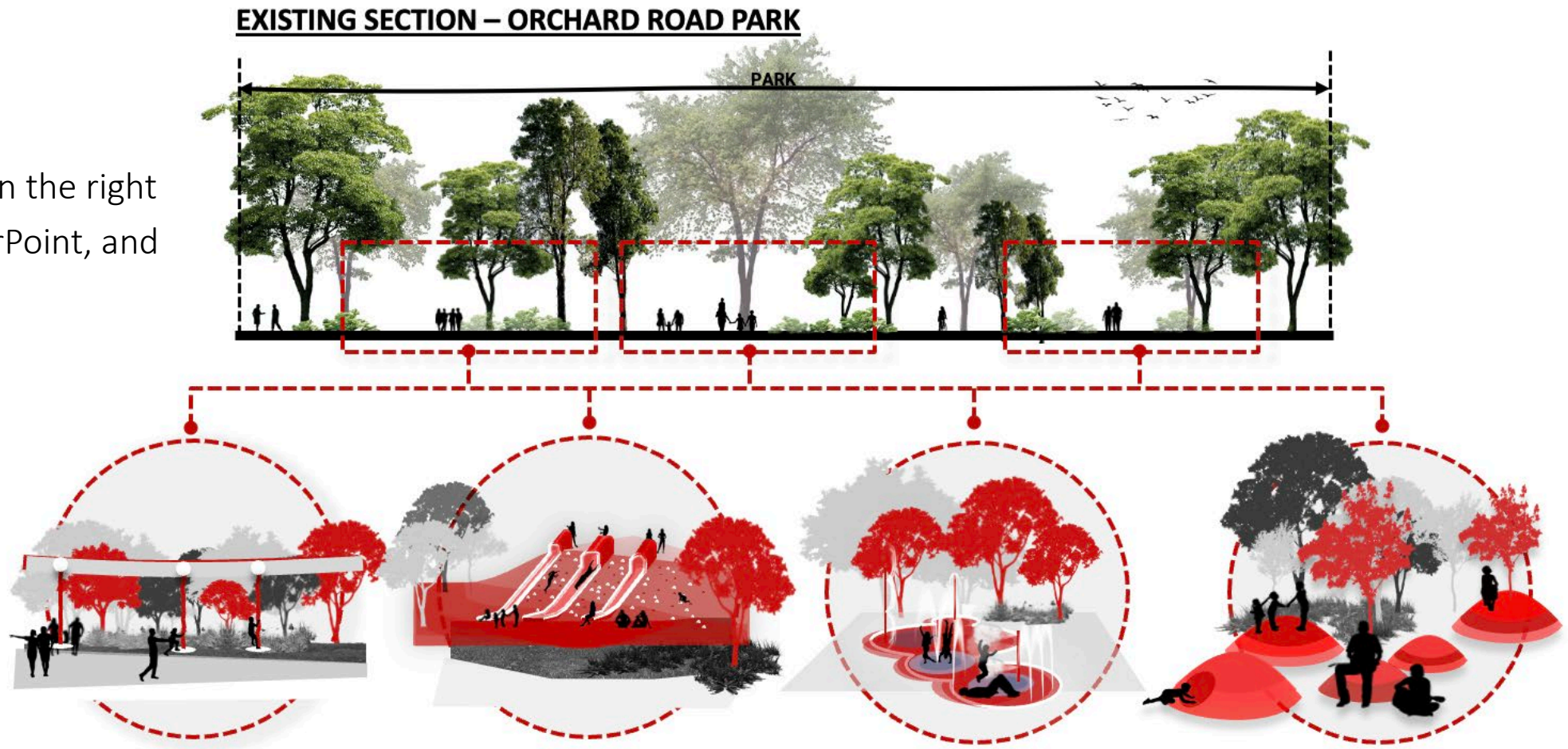


Source: MSc ISD studio, Feb 2021, Jurong
Taken from: Design Communication – Representing the Site

Visualization for Proposal Concepts

Photo Collages

Note: The graphic on the right was made on PowerPoint, and no fancy software!



Source: EG2501 Liveable Cities, Central Area 5 Team 1, Semester 1 AY2022-23

Visualization for Proposal Concepts

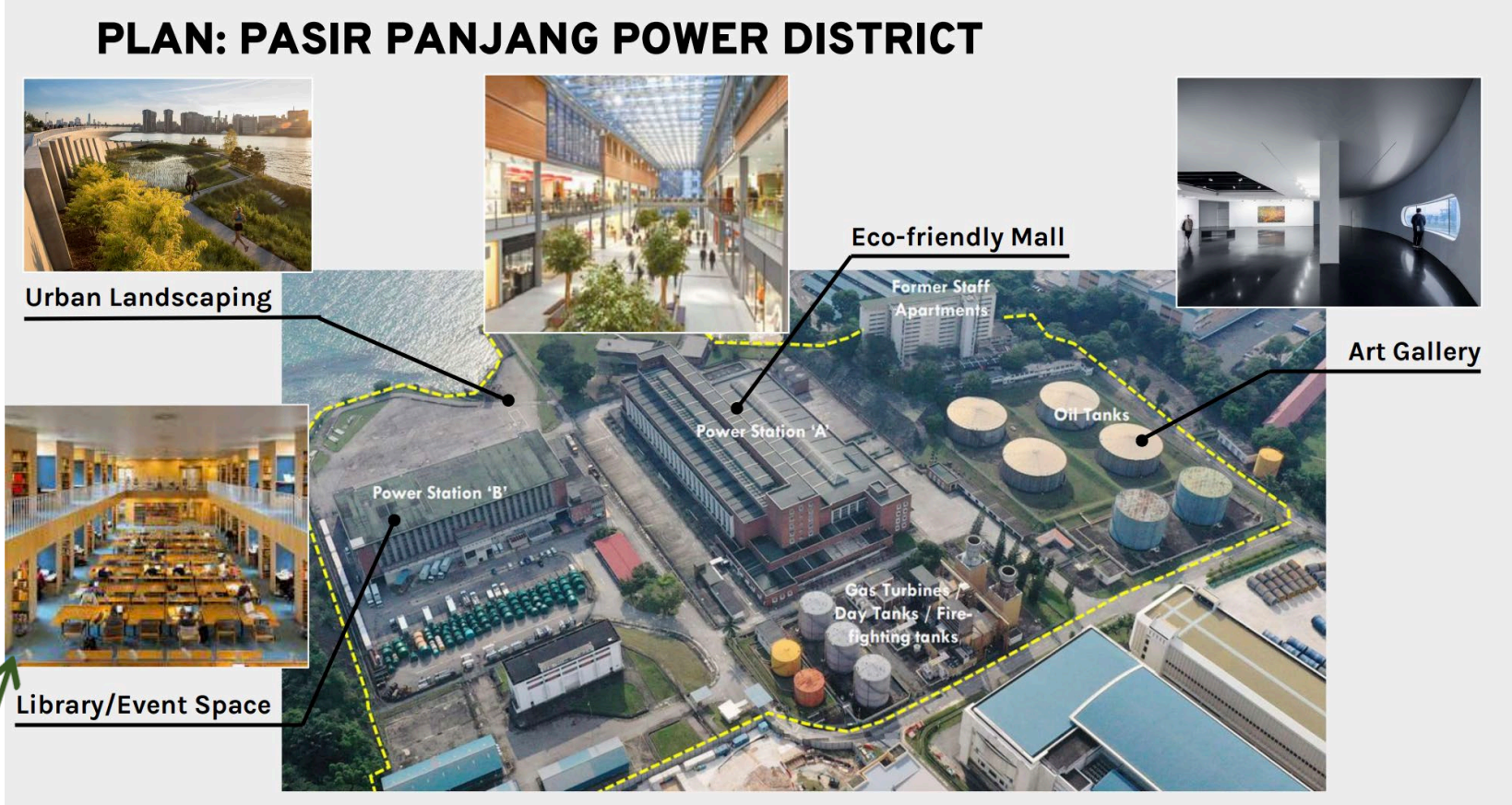
HIGHLIGHT KEY AREAS

Using photos and words to highlight area of interests/solutions in your district.

Remember to give context or background information to your photos too.

**Remember not to overcomplicate your montage!*

Case studies will be useful as ready resource of similar concepts for mood boards



Source: EG2501 Liveable Cities, Queenstown 5 Team 1 Semester 1, 2022-23

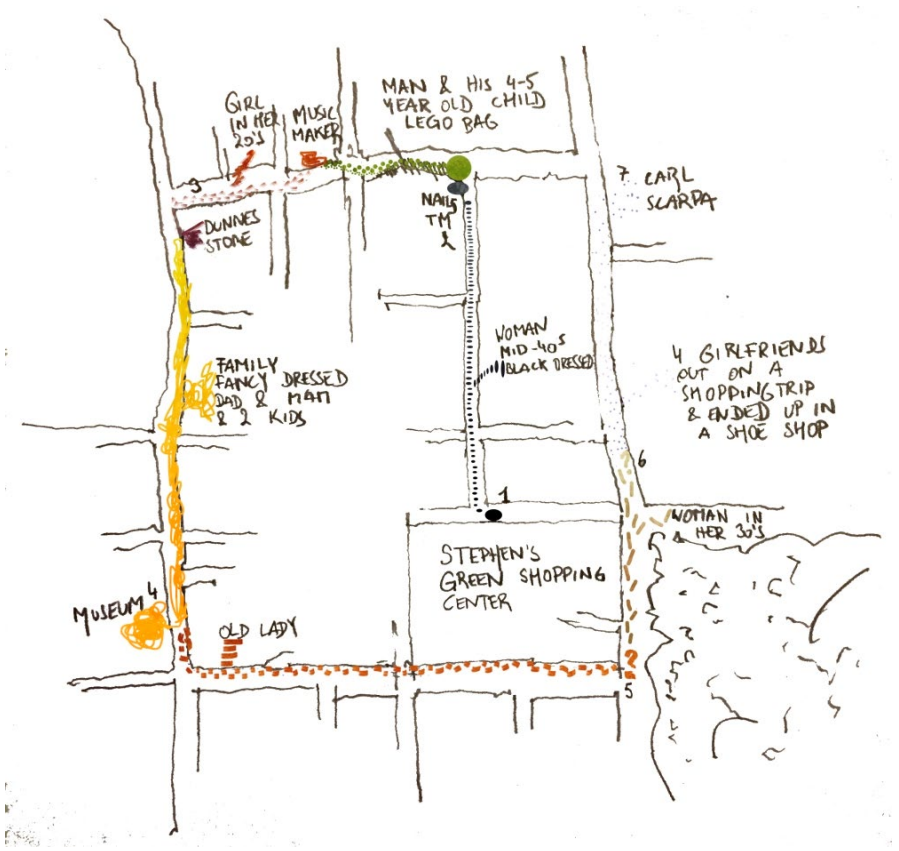
Visualization for Proposal Concepts

SIMPLE SKETCHES

Simple hand draft sketches can also be used to convey ideas and concepts!

Like the two examples in this slide, it is always good to use different colors for easier visualization.

**Remember to include descriptions to your sketches to give context to your audience!*



Source: https://architectureandtime.wordpress.com/situationist-mapping-exercise/

Mapping human traffic

Source: Central Area District 6 Team 5 AY2022/23 Sem 1

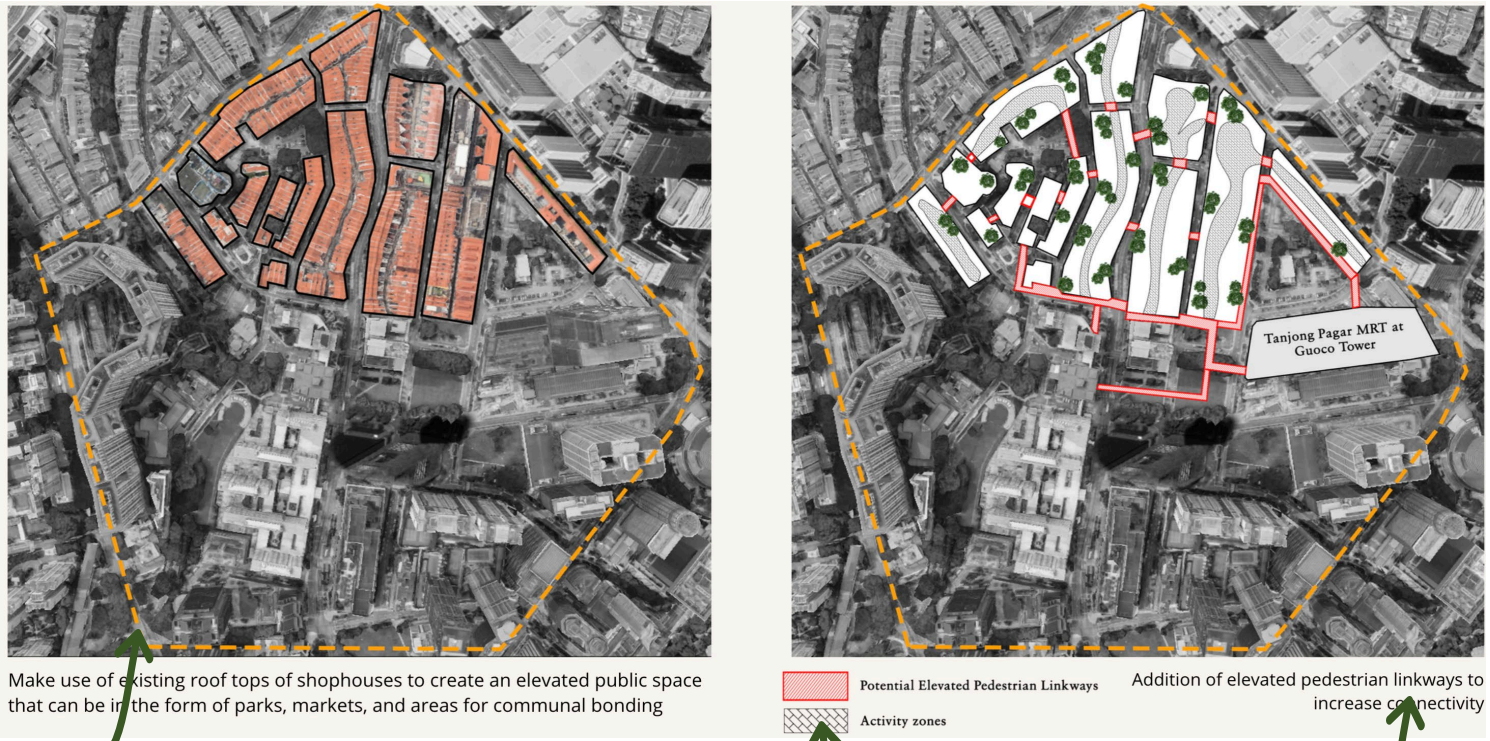
Basic sketches on top of photos

Visualization for Proposal Concepts

MAPS

Using maps online, you can combine your sketches or photos to visualize your ideas. Helps to spatially orientate your audience.

**Remember to include descriptions to your sketches to give context to your audience!*



Source: EG2501 Liveable Cities, Central Area 6 Team 1, Semester 1 AY2022-23

Highlight your study boundary and areas of interest within

Legends and titles are critical in maps

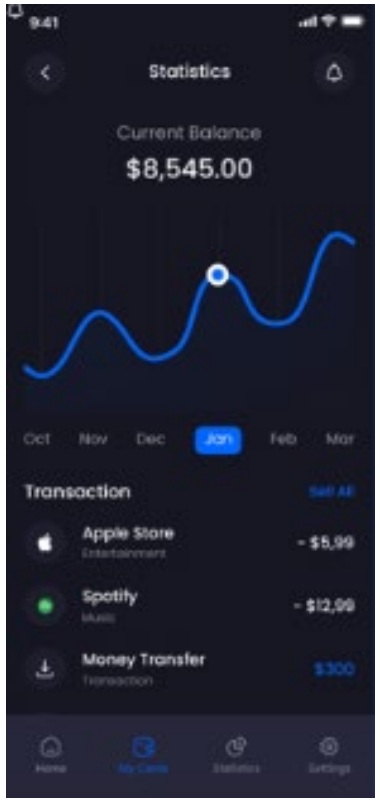
Visualization for Proposal Concepts

BE CREATIVE, HAVE FUN

Remember to annotate

"3D Model" from Central District Team
AY2023/24 Sem 1

Midjourney visualization by Dunman High School students for URA Challenge for the Urban and Built Environment (CUBE) competition. Source: URA

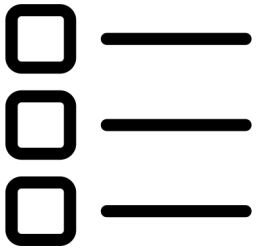


Figma app prototypes
Source: Freebiesbug

Creating an Effective Presentation

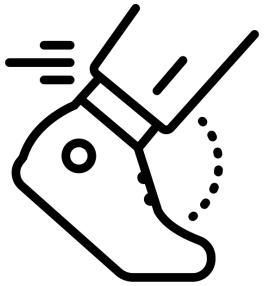
Creating an Effective Presentation

PRESENTATION



Core Objective, Be succinct

- What are the most important pieces of information that also cover the deliverables?
- What are the key takeaways?
- What data is there to back it up?

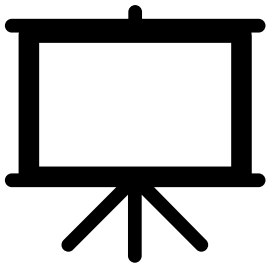


Pace yourself

- Allow for enough time to explain each component while covering all important information? REHEARSE as a GROUP!
- How could one condense the information?
    - For Example: Show most salient tradeoffs not the process, focus on crucial aspects to contextualize the district

NUS CITIES

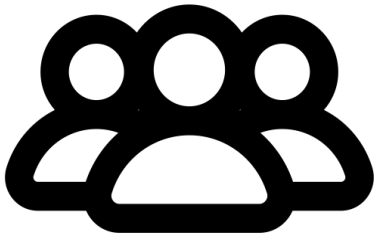
Creating an Effective Presentation



Slide Curation

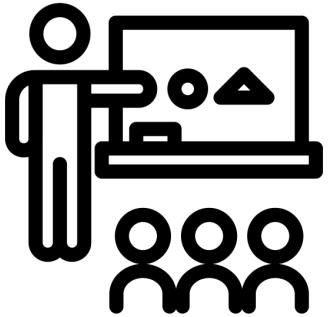
- Curate the text and graphics in each slides – ensure its not too heavy
- Be very careful and smart with the way you use color, line thickness and font sizes
- Focus on a few points per slides – no need to explain everything – it can become overwhelming sometimes
- Referencing of images, data and information – very important and crucial
- Use the appendix to store relevant and supplementary information that does not fit within the ten-minute presentation – as well as some of your old and relevant slides

Creating an Effective Presentation



Gauging the Audience's Needs

- Guest reviewers will not have viewed your projects beforehand
- What pieces of information are necessary for them to understand your project? What do they need to know?
    - Tip: Write down the most crucial points per slides in your speaker notes



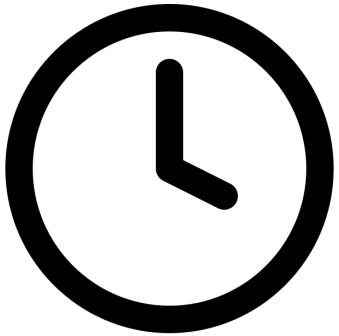
Engagement with your audience

- How can you make the presentation more compelling? What is the most effective way to convey information?
- What will gain the audience's attention?
- How will you encourage the audience to engage with the information?
- We are interested when you are interested. Your body language speaks volumes about your interest and engagement with the project
- Avoid narrating from the script – use your script to only remind you of the key points to be covered

Creating an Effective Presentation

Use Teamwork

- Rehearse as a team, not as individuals – since you need to understand how each person speaks as well as tune the pace of speaking
- Know ALL component parts of the project – THIS IS YOUR GROUP PROJECT



Keep to the Time

- Stick to 10 minutes and rehearse if needed
- Outline the flow of your presentation beforehand
- You will be stopped at the 10-minute mark